

MAIE 5421 COMPUTER VISION: ASSIGNMENT-1

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Question 1: Simple Linear Regression

Given the simple linear regression model: $Y = \alpha + \beta X + \epsilon$, we derive the parameter estimates using the method of least squares.

The objective is to minimize the sum of squared residuals:

$$L = \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2$$

Taking partial derivatives with respect to α and β :

$$\frac{\partial L}{\partial \alpha} = -2 \sum_{i=1}^n (Y_i - \alpha - \beta X_i) = 0$$

$$\frac{\partial L}{\partial \beta} = -2 \sum_{i=1}^n X_i (Y_i - \alpha - \beta X_i) = 0$$

Solving these equations simultaneously:

From the first equation:

$$\sum Y_i - n\alpha - \beta \sum X_i = 0 \Rightarrow \alpha = \bar{Y} - \beta \bar{X}$$

Substituting into the second equation:

$$\begin{aligned} \sum X_i Y_i - \alpha \sum X_i - \beta \sum X_i^2 &= 0 \\ \sum X_i Y_i - (\bar{Y} - \beta \bar{X}) \sum X_i - \beta \sum X_i^2 &= 0 \end{aligned}$$

Solving for β :

$$\beta = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

Thus, the parameter estimates are:

$$\hat{\beta} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}, \quad \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

Question 2: Confusion Matrix

Given the confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	TP = 40	FN = 10
Actual Negative	FP = 5	TN = 45

The evaluation metrics are calculated as follows:

$$\text{True Positive Rate (TPR/Recall)} = \frac{40}{40 + 10} = 0.8$$

$$\text{True Negative Rate (TNR)} = \frac{45}{45 + 5} = 0.9$$

$$\text{False Positive Rate (FPR)} = \frac{5}{45 + 5} = 0.1$$

$$\text{False Negative Rate (FNR)} = \frac{10}{40 + 10} = 0.2$$

$$\text{Precision} = \frac{40}{40 + 5} = \frac{40}{45} \approx 0.889$$

$$\text{Accuracy} = \frac{40 + 45}{100} = 0.85$$

$$\text{F1-Score} = 2 \times \frac{0.889 \times 0.8}{0.889 + 0.8} \approx 0.842$$

Question 3: K-Nearest Neighbors

Given dataset from Table 2 and test point (3.5, 4.0, 6.0) with $K = 3$.

L1 Distance Calculation:

For each point $P_i = (x_i, y_i, z_i)$ in the dataset:

$$d_{L1}(P_{\text{test}}, P_i) = |3.5 - x_i| + |4.0 - y_i| + |6.0 - z_i|$$

$$d_{L1}(P_{\text{test}}, P_1) = |3.5 - 1.0| + |4.0 - 2.0| + |6.0 - 3.0| = 2.5 + 2.0 + 3.0 = 7.5$$

$$d_{L1}(P_{\text{test}}, P_2) = |3.5 - 0.5| + |4.0 - 1.8| + |6.0 - 2.7| = 3.0 + 2.2 + 3.3 = 8.5$$

$$d_{L1}(P_{\text{test}}, P_3) = |3.5 - 1.2| + |4.0 - 2.2| + |6.0 - 3.5| = 2.3 + 1.8 + 2.5 = 6.6$$

$$d_{L1}(P_{\text{test}}, P_4) = |3.5 - 4.6| + |4.0 - 5.6| + |6.0 - 3.7| = 1.1 + 1.6 + 2.3 = 5.0$$

$$d_{L1}(P_{\text{test}}, P_5) = |3.5 - 2.4| + |4.0 - 4.6| + |6.0 - 3.6| = 1.1 + 0.6 + 2.4 = 4.1$$

$$d_{L1}(P_{\text{test}}, P_6) = |3.5 - 3.5| + |4.0 - 2.0| + |6.0 - 4.1| = 0.0 + 2.0 + 1.9 = 3.9$$

$$d_{L1}(P_{\text{test}}, P_7) = |3.5 - 3.6| + |4.0 - 4.6| + |6.0 - 7.1| = 0.1 + 0.6 + 1.1 = 1.8$$

$$d_{L1}(P_{\text{test}}, P_8) = |3.5 - 6.2| + |4.0 - 4.1| + |6.0 - 1.3| = 2.7 + 0.1 + 4.7 = 7.5$$

$$d_{L1}(P_{\text{test}}, P_9) = |3.5 - 8.4| + |4.0 - 3.5| + |6.0 - 1.8| = 4.9 + 0.5 + 4.2 = 9.6$$

$$d_{L1}(P_{\text{test}}, P_{10}) = |3.5 - 5.8| + |4.0 - 3.4| + |6.0 - 2.7| = 2.3 + 0.6 + 3.3 = 6.2$$

Point 7: $d = 1.8$ (Label A), Point 6: $d = 3.9$ (Label B), Point 5: $d = 4.1$ (Label A)
3 nearest neighbors are points 7, 6, 5 with labels A, B, A. Majority vote: Class A.

Cosine Distance Calculation:

$$\text{Similarity} = \frac{P_{\text{test}} \cdot P_i}{\|P_{\text{test}}\| \|P_i\|} = \frac{3.5x_i + 4.0y_i + 6.0z_i}{\sqrt{3.5^2 + 4.0^2 + 6.0^2} \cdot \sqrt{x_i^2 + y_i^2 + z_i^2}}$$

$$\text{Cosine Distance} = 1 - \text{Similarity}$$

$$\|P_{\text{test}}\| = \sqrt{3.5^2 + 4.0^2 + 6.0^2} = \sqrt{12.25 + 16 + 36} = \sqrt{64.25} \approx 8.016$$

$$\|P_1\| = \sqrt{1.0^2 + 2.0^2 + 3.0^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \approx 3.742$$

$$\text{Similarity}(P_{\text{test}}, P_1) = \frac{3.5 \times 1.0 + 4.0 \times 2.0 + 6.0 \times 3.0}{8.016 \times 3.742} = \frac{3.5 + 8.0 + 18.0}{29.99} = \frac{29.5}{29.99} \approx 0.9837$$

$$\text{Cosine Distance}(P_1) = 1 - 0.9837 = 0.0163$$

$$\|P_2\| = \sqrt{0.5^2 + 1.8^2 + 2.7^2} = \sqrt{0.25 + 3.24 + 7.29} = \sqrt{10.78} \approx 3.284$$

$$\text{Similarity}(P_{\text{test}}, P_2) = \frac{3.5 \times 0.5 + 4.0 \times 1.8 + 6.0 \times 2.7}{8.016 \times 3.284} = \frac{1.75 + 7.2 + 16.2}{26.33} = \frac{25.15}{26.33} \approx 0.9552$$

$$\text{Cosine Distance}(P_2) = 1 - 0.9552 = 0.0448$$

$$\begin{aligned}
\|P_3\| &= \sqrt{1.2^2 + 2.2^2 + 3.5^2} = \sqrt{1.44 + 4.84 + 12.25} = \sqrt{18.53} \approx 4.305 \\
\text{Similarity}(P_{\text{test}}, P_3) &= \frac{3.5 \times 1.2 + 4.0 \times 2.2 + 6.0 \times 3.5}{8.016 \times 4.305} = \frac{4.2 + 8.8 + 21.0}{34.51} = \frac{34.0}{34.51} \approx 0.9852 \\
\text{Cosine Distance}(P_3) &= 1 - 0.9852 = 0.0148 \\
\|P_4\| &= \sqrt{4.6^2 + 5.6^2 + 3.7^2} = \sqrt{21.16 + 31.36 + 13.69} = \sqrt{66.21} \approx 8.137 \\
\text{Similarity}(P_{\text{test}}, P_4) &= \frac{3.5 \times 4.6 + 4.0 \times 5.6 + 6.0 \times 3.7}{8.016 \times 8.137} = \frac{16.1 + 22.4 + 22.2}{65.22} = \frac{60.7}{65.22} \approx 0.9307 \\
\text{Cosine Distance}(P_4) &= 1 - 0.9307 = 0.0693 \\
\|P_5\| &= \sqrt{2.4^2 + 4.6^2 + 3.6^2} = \sqrt{5.76 + 21.16 + 12.96} = \sqrt{39.88} \approx 6.315 \\
\text{Similarity}(P_{\text{test}}, P_5) &= \frac{3.5 \times 2.4 + 4.0 \times 4.6 + 6.0 \times 3.6}{8.016 \times 6.315} = \frac{8.4 + 18.4 + 21.6}{50.61} = \frac{48.4}{50.61} \approx 0.9563 \\
\text{Cosine Distance}(P_5) &= 1 - 0.9563 = 0.0437 \\
\|P_6\| &= \sqrt{3.5^2 + 2.0^2 + 4.1^2} = \sqrt{12.25 + 4.0 + 16.81} = \sqrt{33.06} \approx 5.750 \\
\text{Similarity}(P_{\text{test}}, P_6) &= \frac{3.5 \times 3.5 + 4.0 \times 2.0 + 6.0 \times 4.1}{8.016 \times 5.750} = \frac{12.25 + 8.0 + 24.6}{46.09} = \frac{44.85}{46.09} \approx 0.9731 \\
\text{Cosine Distance}(P_6) &= 1 - 0.9731 = 0.0269 \\
\|P_7\| &= \sqrt{3.6^2 + 4.6^2 + 7.1^2} = \sqrt{12.96 + 21.16 + 50.41} = \sqrt{84.53} \approx 9.194 \\
\text{Similarity}(P_{\text{test}}, P_7) &= \frac{3.5 \times 3.6 + 4.0 \times 4.6 + 6.0 \times 7.1}{8.016 \times 9.194} = \frac{12.6 + 18.4 + 42.6}{73.70} = \frac{73.6}{73.70} \approx 0.9986 \\
\text{Cosine Distance}(P_7) &= 1 - 0.9986 = 0.0014 \\
\|P_8\| &= \sqrt{6.2^2 + 4.1^2 + 1.3^2} = \sqrt{38.44 + 16.81 + 1.69} = \sqrt{56.94} \approx 7.546 \\
\text{Similarity}(P_{\text{test}}, P_8) &= \frac{3.5 \times 6.2 + 4.0 \times 4.1 + 6.0 \times 1.3}{8.016 \times 7.546} = \frac{21.7 + 16.4 + 7.8}{60.49} = \frac{45.9}{60.49} \approx 0.7588 \\
\text{Cosine Distance}(P_8) &= 1 - 0.7588 = 0.2412 \\
\|P_9\| &= \sqrt{8.4^2 + 3.5^2 + 1.8^2} = \sqrt{70.56 + 12.25 + 3.24} = \sqrt{86.05} \approx 9.276 \\
\text{Similarity}(P_{\text{test}}, P_9) &= \frac{3.5 \times 8.4 + 4.0 \times 3.5 + 6.0 \times 1.8}{8.016 \times 9.276} = \frac{29.4 + 14.0 + 10.8}{74.36} = \frac{54.2}{74.36} \approx 0.7290 \\
\text{Cosine Distance}(P_9) &= 1 - 0.7290 = 0.2710 \\
\|P_{10}\| &= \sqrt{5.8^2 + 3.4^2 + 2.7^2} = \sqrt{33.64 + 11.56 + 7.29} = \sqrt{52.49} \approx 7.245 \\
\text{Similarity}(P_{\text{test}}, P_{10}) &= \frac{3.5 \times 5.8 + 4.0 \times 3.4 + 6.0 \times 2.7}{8.016 \times 7.245} = \frac{20.3 + 13.6 + 16.2}{58.08} = \frac{50.1}{58.08} \approx 0.8626 \\
\text{Cosine Distance}(P_{10}) &= 1 - 0.8626 = 0.1374
\end{aligned}$$

Point 7 ($d = 0.0014$, Label A), Point 3 ($d = 0.0148$, Label B), Point 1 ($d = 0.0163$, Label A). 3 nearest neighbors are points 7, 3, 1 with labels A, B, A. Majority vote: Class A.

Question 4: Naive Bayes Classifier

Given training data from Table 3 and new sample: [Sunny, Hot, Normal, False].

Prior Probabilities:

$$P(\text{Play}=\text{Yes}) = \frac{5}{8}, \quad P(\text{Play}=\text{No}) = \frac{3}{8}$$

Conditional Probabilities:

For $\text{Play}=\text{Yes}$:

$$\begin{aligned} P(\text{Sunny}|\text{Yes}) &= \frac{2}{5}, & P(\text{Hot}|\text{Yes}) &= \frac{2}{5}, \\ P(\text{Normal}|\text{Yes}) &= \frac{3}{5}, & P(\text{False}|\text{Yes}) &= \frac{4}{5} \end{aligned}$$

For $\text{Play}=\text{No}$:

$$\begin{aligned} P(\text{Sunny}|\text{No}) &= \frac{1}{3}, & P(\text{Hot}|\text{No}) &= \frac{1}{3}, \\ P(\text{Normal}|\text{No}) &= \frac{1}{3}, & P(\text{False}|\text{No}) &= \frac{2}{3} \end{aligned}$$

Posterior Probabilities:

$$\begin{aligned} P(\text{Yes}|X) &\propto \frac{5}{8} \cdot \frac{2}{5} \cdot \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{240}{5000} = 0.048 \\ P(\text{No}|X) &\propto \frac{3}{8} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{6}{648} \approx 0.00926 \end{aligned}$$

Decision:

Since $P(\text{Yes}|X) > P(\text{No}|X)$, the predicted class is Yes.

Question 5: Decision Tree

Given dataset from Table 4.

Entropy of Entire Dataset

Total samples = 9, Yes = 6, No = 3.

$$H(S) = -\frac{6}{9} \log_2 \left(\frac{6}{9} \right) - \frac{3}{9} \log_2 \left(\frac{3}{9} \right) \approx 0.918$$

Information Gain for Each Feature

Weather Feature:

Sunny: 3 samples (1 Yes, 2 No) $\rightarrow H = -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \approx 0.918$

Overcast: 3 samples (3 Yes, 0 No) $\rightarrow H = 0$

Rain: 3 samples (3 Yes, 0 No) $\rightarrow H = 0$

$$IG(S, \text{Weather}) = 0.918 - \left(\frac{3}{9} \cdot 0.918 + \frac{3}{9} \cdot 0 + \frac{3}{9} \cdot 0 \right) = 0.918 - 0.306 = 0.612$$

Temperature Feature:

High: 2 samples (1 Yes, 1 No) $\rightarrow H = 1.0$

Medium: 5 samples (4 Yes, 1 No) $\rightarrow H = -\frac{4}{5} \log_2 \left(\frac{4}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \approx 0.722$

Low: 2 samples (2 Yes, 0 No) $\rightarrow H = 0$

$$IG(S, \text{Temperature}) = 0.918 - \left(\frac{2}{9} \cdot 1.0 + \frac{5}{9} \cdot 0.722 + \frac{2}{9} \cdot 0 \right) \approx 0.918 - 0.623 = 0.295$$

Wind Feature:

High: 5 samples (3 Yes, 2 No) $\rightarrow H = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \approx 0.971$

Medium: 4 samples (3 Yes, 1 No) $\rightarrow H = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \approx 0.811$

$$IG(S, \text{Wind}) = 0.918 - \left(\frac{5}{9} \cdot 0.971 + \frac{4}{9} \cdot 0.811 \right) \approx 0.918 - 0.900 = 0.018$$

Root Node Selection

Weather has the highest information gain (0.612), so it becomes the root node.

Build Decision Tree Recursively

Weather = Overcast Branch (Pure Node)

All 3 samples have Class = Yes. This becomes a leaf node: Class = Yes.

Weather = Rain Branch (Pure Node)

All 3 samples have Class = Yes. This becomes a leaf node: Class = Yes.

Weather = Sunny Branch

3 samples (1 Yes, 2 No). Need to split further.

Calculate information gain for remaining features (Temperature, Wind) on this subset:

- Entropy of Sunny subset: $H = 0.918$
- Temperature on Sunny subset:
 - High: 1 sample (0 Yes, 1 No) $\rightarrow H = 0$
 - Medium: 2 samples (1 Yes, 1 No) $\rightarrow H = 1.0$
 - Low: 0 samples $\rightarrow$ Ignore
$$IG = 0.918 - \left(\frac{1}{3} \cdot 0 + \frac{2}{3} \cdot 1.0\right) = 0.918 - 0.667 = 0.251$$
- Wind on Sunny subset:
 - High: 2 samples (0 Yes, 2 No) $\rightarrow H = 0$
 - Medium: 1 sample (1 Yes, 0 No) $\rightarrow H = 0$
$$IG = 0.918 - \left(\frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 0\right) = 0.918 - 0 = 0.918$$

Wind has higher IG (0.918), so we choose Wind for the Sunny branch.

Sunny + Wind = High Branch: Pure (2 No) $\rightarrow$ Leaf: Class = No

Sunny + Wind = Medium Branch: Pure (1 Yes) $\rightarrow$ Leaf: Class = Yes

Final Decision Tree

The complete decision tree can be represented as:

```
Weather
|
+-- Overcast --> Class = Yes
|
+-- Rain -----> Class = Yes
|
+-- Sunny
    |
    +-- Wind = High ----> Class = No
    |
    +-- Wind = Medium --> Class = Yes
```

Classification of New Sample

For a new sample with attributes [Sunny, Hot, Normal, False]:

Following the decision tree:

Weather = Sunny $\rightarrow$ go to Sunny branch

Wind = False (equivalent to Normal/Medium) $\rightarrow$ go to Wind = Medium branch

Result: Class = Yes

Question 6: K-Means

Given the 3-dimensional dataset containing 8 data points. Use the K-Means algorithm to divide these data points into 2 clusters ($k = 2$) and calculate the centroid of each cluster.

Initial Centroids

Select 2 initial centroids (Data Point 1 and Data Point 3):

$$C_1 = (2.0, 3.0, 1.0)$$

$$C_2 = (8.0, 7.0, 9.0)$$

Euclidean Distance Calculation

Euclidean distance formula:

$$d(P, Q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Iterative Updates

Round 1: Assign Points to Closest Centroid

Calculate distances and assign points:

Point 1:

$$d(C_1) = 0$$

$$d(C_2) = \sqrt{(2 - 8)^2 + (3 - 7)^2 + (1 - 9)^2} = \sqrt{36 + 16 + 64} = \sqrt{116} \approx 10.77$$

Assign to C_1

Point 2:

$$d(C_1) = \sqrt{(1.5 - 2)^2 + (2.5 - 3)^2 + (1.2 - 1)^2} = \sqrt{0.25 + 0.25 + 0.04} = \sqrt{0.54} \approx 0.73$$

$$d(C_2) \approx 11.27$$

Assign to C_1

Point 3:

$$d(C_1) \approx 10.77$$

$$d(C_2) = 0$$

Assign to C_2

Point 4:

$$d(C_1) \approx 11.22$$

$$d(C_2) = \sqrt{(7.5 - 8)^2 + (6.5 - 7)^2 + (8.5 - 9)^2} = \sqrt{0.25 + 0.25 + 0.25} = \sqrt{0.75} \approx 0.87$$

Assign to C_2

Point 5:

$$d(C_1) = \sqrt{(2.5 - 2)^2 + (3.5 - 3)^2 + (1.5 - 1)^2} = \sqrt{0.25 + 0.25 + 0.25} = \sqrt{0.75} \approx 0.87$$

$$d(C_2) \approx 10.39$$

Assign to C_1

Point 6:

$$d(C_1) \approx 10.05$$

$$d(C_2) = \sqrt{(7 - 8)^2 + (6 - 7)^2 + (8 - 9)^2} = \sqrt{1 + 1 + 1} = \sqrt{3} \approx 1.73$$

Assign to C_2

Point 7:

$$d(C_1) = \sqrt{(1.8 - 2)^2 + (2.8 - 3)^2 + (1.1 - 1)^2} = \sqrt{0.04 + 0.04 + 0.01} = \sqrt{0.09} = 0.3$$

$$d(C_2) \approx 10.95$$

Assign to C_1

Point 8:

$$d(C_1) \approx 11.53$$

$$d(C_2) = \sqrt{(8.5 - 8)^2 + (7.5 - 7)^2 + (9.5 - 9)^2} = \sqrt{0.25 + 0.25 + 0.25} = \sqrt{0.75} \approx 0.87$$

Assign to C_2

Cluster assignments after Round 1:

Cluster 1 (C_1): Points 1, 2, 5, 7

Cluster 2 (C_2): Points 3, 4, 6, 8

Update Centroids after Round 1

Calculate new centroids as the mean of points in each cluster:

$$\text{New } C_1 = \left(\frac{2.0 + 1.5 + 2.5 + 1.8}{4}, \frac{3.0 + 2.5 + 3.5 + 2.8}{4}, \frac{1.0 + 1.2 + 1.5 + 1.1}{4} \right)$$

$$= \left(\frac{7.8}{4}, \frac{11.8}{4}, \frac{4.8}{4} \right) = (1.95, 2.95, 1.2)$$

$$\text{New } C_2 = \left(\frac{8.0 + 7.5 + 7.0 + 8.5}{4}, \frac{7.0 + 6.5 + 6.0 + 7.5}{4}, \frac{9.0 + 8.5 + 8.0 + 9.5}{4} \right)$$

$$= \left(\frac{31.0}{4}, \frac{27.0}{4}, \frac{35.0}{4} \right) = (7.75, 6.75, 8.75)$$

Round 2: Assign Points to Closest Centroid

Calculate distances to new centroids and assign points:

Point 1:

$$d(C_1) = \sqrt{(2 - 1.95)^2 + (3 - 2.95)^2 + (1 - 1.2)^2} = \sqrt{0.0025 + 0.0025 + 0.04} = \sqrt{0.045} \approx 0.21$$

$$d(C_2) = \sqrt{(2 - 7.75)^2 + (3 - 6.75)^2 + (1 - 8.75)^2} = \sqrt{33.0625 + 14.0625 + 60.0625} = \sqrt{107.1875} \approx 10.35$$

Assign to C_1

Point 2:

$$d(C_1) = \sqrt{(1.5 - 1.95)^2 + (2.5 - 2.95)^2 + (1.2 - 1.2)^2} = \sqrt{0.2025 + 0.2025 + 0} = \sqrt{0.405} \approx 0.64$$

$$d(C_2) \approx 10.79$$

Assign to C_1

Point 3:

$$d(C_1) \approx 10.64$$

$$d(C_2) = \sqrt{(8 - 7.75)^2 + (7 - 6.75)^2 + (9 - 8.75)^2} = \sqrt{0.0625 + 0.0625 + 0.0625} = \sqrt{0.1875} \approx 0.43$$

Assign to C_2

Point 4:

$$d(C_1) \approx 11.06$$

$$d(C_2) = \sqrt{(7.5 - 7.75)^2 + (6.5 - 6.75)^2 + (8.5 - 8.75)^2} = \sqrt{0.0625 + 0.0625 + 0.0625} = \sqrt{0.1875} \approx 0.43$$

Assign to C_2

Point 5:

$$d(C_1) = \sqrt{(2.5 - 1.95)^2 + (3.5 - 2.95)^2 + (1.5 - 1.2)^2} = \sqrt{0.3025 + 0.3025 + 0.09} = \sqrt{0.695} \approx 0.83$$

$$d(C_2) \approx 9.97$$

Assign to C_1

Point 6:

$$d(C_1) \approx 9.83$$

$$d(C_2) = \sqrt{(7 - 7.75)^2 + (6 - 6.75)^2 + (8 - 8.75)^2} = \sqrt{0.5625 + 0.5625 + 0.5625} = \sqrt{1.6875} \approx 1.30$$

Assign to C_2

Point 7:

$$d(C_1) = \sqrt{(1.8 - 1.95)^2 + (2.8 - 2.95)^2 + (1.1 - 1.2)^2} = \sqrt{0.0225 + 0.0225 + 0.01} = \sqrt{0.055} \approx 0.23$$

$$d(C_2) \approx 10.77$$

Assign to C_1

Point 8:

$$d(C_1) \approx 11.34$$

$$d(C_2) = \sqrt{(8.5 - 7.75)^2 + (7.5 - 6.75)^2 + (9.5 - 8.75)^2} = \sqrt{0.5625 + 0.5625 + 0.5625} = \sqrt{1.6875} \approx 1.3$$

Assign to C_2

Cluster assignments after Round 2:

Cluster 1 (C_1): Points 1, 2, 5, 7 (no change from Round 1)

Cluster 2 (C_2): Points 3, 4, 6, 8 (no change from Round 1)

Since cluster assignments did not change, the centroids remain the same as after Round 1.

Final Output

- **Final Centroid 1:** (1.95, 2.95, 1.2)
- **Points in Cluster 1:** 1, 2, 5, 7
- **Final Centroid 2:** (7.75, 6.75, 8.75)
- **Points in Cluster 2:** 3, 4, 6, 8

Question 7: CNN Q1

Calculate total parameters in the CNN architecture.

Layer-by-Layer Calculation:

1. **Conv1:** Input: $1 \times 64 \times 64$, Output: $8 \times 32 \times 32$

$$\text{Parameters} = (3 \times 3 \times 1 + 1) \times 8 = 10 \times 8 = 80$$

2. **Conv2:** Input: $8 \times 32 \times 32$, Output: $16 \times 16 \times 16$

$$\text{Parameters} = (3 \times 3 \times 8 + 1) \times 16 = 73 \times 16 = 1168$$

3. **Conv3:** Input: $16 \times 16 \times 16$, Output: $32 \times 16 \times 16$

$$\text{Parameters} = (3 \times 3 \times 16 + 1) \times 32 = 145 \times 32 = 4640$$

4. **Max Pooling:** No parameters

5. **Flatten:** No parameters

6. **Linear Layer:** Input: 2048, Output: 10

$$\text{Parameters} = (2048 \times 10) + 10 = 20480 + 10 = 20490$$

Total Parameters:

$$80 + 1168 + 4640 + 20490 = 26378$$

Question 8: CNN Q2

Given: Input matrix A (5×5), Kernel K (3×3), Weight matrix W (9×1).

Convolution Operation

No padding, stride=1. Output size: $(5 - 3 + 1) \times (5 - 3 + 1) = 3 \times 3$.

Calculate convolution at each position:

$$\text{Position (0,0): } A[0 : 3, 0 : 3] \cdot K = -17$$

$$\text{Position (0,1): } A[0 : 3, 1 : 4] \cdot K = 6$$

$$\text{Position (0,2): } A[0 : 3, 2 : 5] \cdot K = 18$$

$$\text{Position (1,0): } A[1 : 4, 0 : 3] \cdot K = 3$$

$$\text{Position (1,1): } A[1 : 4, 1 : 4] \cdot K = 1$$

$$\text{Position (1,2): } A[1 : 4, 2 : 5] \cdot K = -1$$

$$\text{Position (2,0): } A[2 : 5, 0 : 3] \cdot K = 4$$

$$\text{Position (2,1): } A[2 : 5, 1 : 4] \cdot K = -8$$

$$\text{Position (2,2): } A[2 : 5, 2 : 5] \cdot K = -5$$

Convolution result:

$$C = \begin{bmatrix} -17 & 6 & 18 \\ 3 & 1 & -1 \\ 4 & -8 & -5 \end{bmatrix}$$

ReLU Activation

Apply ReLU function: $\text{ReLU}(x) = \max(0, x)$

$$C_{\text{ReLU}} = \begin{bmatrix} \max(0, -17) & \max(0, 6) & \max(0, 18) \\ \max(0, 3) & \max(0, 1) & \max(0, -1) \\ \max(0, 4) & \max(0, -8) & \max(0, -5) \end{bmatrix} = \begin{bmatrix} 0 & 6 & 18 \\ 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

Flatten

Convert 3×3 matrix to 9×1 vector:

$$V = [0 \ 6 \ 18 \ 3 \ 1 \ 0 \ 4 \ 0 \ 0]^T$$

Linear Transformation

Multiply by weight matrix W (no bias):

$$\text{Output} = W^T \cdot V = [1 \ -2 \ 1 \ -1 \ -3 \ 0 \ 1 \ -1 \ 1] \cdot [0 \ 6 \ 18 \ 3 \ 1 \ 0 \ 4 \ 0 \ 0]^T = 4$$